Estimating the supply elasticity of innovation

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Abstract

Government R&D may pass through to higher scientists' wages rather than an increase in innovation when the labor supply of scientific workforce is inelastic. This paper leverages granular data on scientists' productivity, wages, and research funding to estimate the impact of government R&D on both inventive output and wages. To identify these effects, I exploit variation in city size driven by universities' historical ties to defense funding, which influenced the geographic distribution of scientists through the placement of graduates into research-intensive cities. I find that government R&D funding increases inventive output, particularly among scientists who do not receive funding themselves. The monetary value of increased inventions outweighs the increase in labor costs. While R&D subsidies do lead to higher wages, the results indicate that government investment in research can enhance overall innovation output through the entry of new scientists and subsequent knowledge spillovers, and that the elasticity of innovation with respect to government funding is large at least in the long run.

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1. Introduction

A key challenge in evaluating the impact of government R&D investment is distinguishing between its effects on innovation and the labor costs of R&D. If the supply of scientists is constrained, increased R&D funding may bid up wages without significantly boosting inventive output. The labor supply curve for scientists is in general considered inelastic due to the substantial costs and time required for training. Training a scientist involves years of specialized education and research experience, limiting the short-run responsiveness of labor supply to increased demand (Goolsbee, 1998; Romer, 2000). Does government R&D primarily stimulate innovation, or does it mainly drive up wages due to the inelastic supply of scientists? To the extent that the effect of government R&D on productivity growth is neither uniform across nations nor constant over time, understanding the interplay between innovation and labor costs offers invaluable insights into the role of government research in shaping economic outcomes.

This paper uses granular data on scientists' productivity, wages, and research funding to jointly investigate the impact of government R&D on both inventive output and wages. Building on Goolsbee (1998), which finds that government R&D raises scientists' wages without increasing working hours, this study presents three key findings that remained unanswered. First, the labor supply of scientists is quite elastic in the long run. Second, government R&D has positive spillover effects on the productivity of scientists who do not receive the funding directly. Third, while government R&D does lead to higher wages, the monetary value of increased invention outweighs the increased labor costs. Taken together, these results suggest that the supply elasticity of innovation with resepect to government funding can be quite large in the long run despite pass through to the labor costs.

To estimate the external effects of government funding on research output and wages, I estimate the geographic knowledge spillover by exploiting variation in city size driven by post-WWII defense funding across universities, interacted with universities' historical propensity to send graduates to different cities. This proposed shift-share instrument is uncorrelated with pre-period patenting activity but strongly predicts city size, allowing me to isolate the causal effect of R&D funding from other confounding factors that simultaneously influence innovation and wages.

Using this empirical strategy, I find that defense funding significantly increases scientists' inventive output while also raising wages. Notably, the wage effects are similar regardless of whether a scientist receives defense funding. However, the increase in patenting activity is observed only among scientists who did not receive direct funding, suggesting that knowledge spillovers from agglomeration primarily benefit non-funded researchers. These findings provide new evidence that government R&D enhances overall innovation rather than merely redistributing resources among existing researchers, underscoring its broader impact beyond direct funding recipients.

Related Literature. This paper contributes to several strands of interrelated literature by providing new insights into the channels through which government R&D funding affects both innovation and wages. First, prior work on the supply elasticity of innovation has examined whether it moderates the responsiveness of innovative output to government R&D (Goolsbee, 1998; David & Hall, 2000; Romer, 2000). Similarly, studies estimating the labor supply elasticity of high-skilled professionals have focused on wage responses to shifts in demand (Goolsbee & Syverson, 2019; Fernández-Villaverde *et al.*, 2024). In contrast, my paper utilizes granular data on scientists' productivity, wages, and funding to separately identify the direct wage effects and the indirect productivity spillovers of government R&D. By distinguishing between funded and non-funded scientists, I demonstrate that although wage increases occur across the board, the boost in inventive output is concentrated among those who do not receive direct funding. This finding deepens our understanding of innovation supply elasticity and highlights the critical role of external knowledge spillovers in the innovation process.

Second, this study introduces a novel approach that leverages rich, individual-level data on scientists to estimate the returns to government R&D (Myers & Lanahan, 2022; Dyevre, 2023; Fieldhouse & Mertens, 2023; Gross & Sampat, 2023; Kantor & Whalley, 2023; Moretti *et al.*, 2025). This method builds on insights from the literature on agglomeration and innovation (Carlino *et al.*, 2007; Burchardi *et al.*, 2020; Moretti, 2021; Gruber *et al.*, 2023; Giroud *et al.*, 2024). The pass-through to wages serves as a crucial parameter for comparing the estimated effects of R&D on productivity across countries and over time. The idea that R&D investment does not necessarily lead to productivity growth due to the monopsonistic labor market is also reflected in the recent study by Fernández-Villaverde *et al.* (2024).

2. Data

I use the National Register of Scientific and Technical Personnel Files from the National Archives (Record Group 307: Records of the National Science Foundation) to obtain individual-level data on scientists in 1970. This dataset provides a comprehensive sample available from these surveys, covering a wide range of scientific disciplines. The survey was conducted in collaboration with professional organizations, including the American Chemical Society, the American Mathematical Society, the American Psychological Association, and the American Institute of Physics, among others. These organizations distributed questionnaires to academic and research professionals across various scientific and technical fields.

The dataset contains detailed demographic and employment information, including salary, educational background, institutional affiliation, research specialization, job function, and government sponsorship. Additional variables include age, sex, citizenship, professional organization memberships, and geographic location (city and state). This dataset allows for a granular examination of wage and the impact of government R&D funding on scientific output. For my analysis, I focus on 276,611 scientists who earned their highest degree between 1946 and 1970 across 13 fields, 1,671 institutions, and 433 Standard Metropolitan Statistical Areas (SMSAs), which I refer to as cities. In Table A.2, I document the correlation between scientists' wages and whether they receive defense funding. The results show that scientists who receive defense funding earn higher wages, even after controlling for institution, graduation year, field, gender, and city fixed effects.

This dataset has been used in recent studies (Kantor & Whalley, 2023; Gross & Roche, 2024). Compared to the academic database in Iaria *et al.* (2018), it has a shorter time coverage but includes scientists from a broader range of employers, such as private industry and nonprofits. Compared with the Men of Science dataset used in Kim & Moser (2021) and Arora *et al.* (2024), it covers a later period and has a larger sample size (300,000 in the National Register vs. 90,000 in Men of Science).

To measure the inventive output of scientists, I use patent value estimates from Kelly *et al.* (2021), which are based on textual similarity. Specifically, I identify important patents by analyzing their textual similarity to both preceding and subsequent innovations. These patents are distinct from earlier work but closely related to later advancements. I calculate patent value by measuring the similarity of a given patent to subsequent patents over a 20-year period, normalized by its similarity to patents from the preceding 5 years. I classify the top 10 percent of patents based on this measure as high-impact patents. This approach accounts for the highly skewed distribution of patent values (Hall, 2000; Kogan *et al.*, 2017; Argente *et al.*, 2020) and also improves computational efficiency when extracting patent author information from Google Patents.

For the main analysis, I focus on patents granted between 1951 and 1976 to capture the inventive output of scientists registered in the 1970 NSF survey. Two potential issues with this step are, first, that it does not account for scientists who relocated across cities over time, and second, the presence of false positives due to incomplete disambiguation in the historical patent data. These issues will be addressed in a later version of this paper.

3. Empirical Strategy

To analyze the external impact of defense funding on inventive output and wages, I estimate the effect of agglomeration using the following equation:

$$y_{icfn} = \beta \ln(POP_{c,-n}) + X_i'\gamma + \varepsilon_{icfn} \tag{1}$$

where y_{icfn} represents an outcome variable, such as the number of patents or wages of individual i in city c, field f, and institution n. The term $POP_{c,-n}$ denotes the size of the scientific workforce in city c, excluding scientists who graduated from the same institution n of the focal scientist, to avoid mechanical correlations driven by the presence of the focal scientist's alumni network. The vector X_i includes individual characteristics such as field of study, institution, graduation year, and a female fixed effect. The error term ε_{icfn} are clustered by city.

The estimated coefficient β is biased if city size and the error term are correlated. This issue arises when unobserved factors that influence the outcome variable y_{icfn} are also correlated with city size. If such factors are omitted from the regression, the estimated coefficient will capture both the effect of city size and the influence of these unobserved factors, leading to biased results.

If larger cities tend to have more favorable unobserved characteristics—such as stronger professional networks, better infrastructure, or greater innovation spillovers—that positively affect individual outcomes, then city size is positively correlated with the error term ($\mathrm{Cov}(\ln(POP_{c,-n}), \varepsilon_{icfn}) > 0$). In this case, the estimated β overstates the true effect of city size. Conversely, if larger cities are associated with unobserved factors that negatively affect outcomes—such as congestion, higher living costs, or increased competition—then city size is negatively correlated with the error term ($\mathrm{Cov}(\ln(POP_{c,-n}), \varepsilon_{icfn}) < 0$). This would lead to an underestimate of the true effect of city size.

Instrumental variable strategy. To address this potential endogeneity issue, an instrumental variable approach or additional control variables capturing local economic conditions may help isolate the causal effect of city size. I construct an instrumental variable based on historical patterns of defense-related funding for scientific training. Specifically, I use the following measure:

$$\sum_{n' \neq n} \Delta F_{n'}(t) \frac{G_{n',c}(t-1)}{\sum_{c'} G_{n',c'}(t-1)} \tag{2}$$

where $G_{n,c}(t-1)$ represents the number of graduates from institution n located in city c before 1945. The term $\Delta F_n(t)$ denotes the number of graduates from institution n who received funding from the Department of Defense between 1946 and 1970.

This measure captures exogenous variation in city size stemming from differential exposure to postwar defense funding, which influenced the geographic distribution of scientists independently of contemporary local economic conditions.

The identification strategy relies on the historical tendency of universities to shape the geographic distribution of scientists. In particular, certain universities have traditionally supplied a substantial share of scientists to specific cities, implying that the location of scientific talent is partly determined by the historical linkages between universities and cities. By exploiting the interaction between this propensity and the variation in postwar defense funding across universities, this measure isolates shifts in the size of the scientific workforce that are driven not by local economic conditions, but by exogenous funding decisions. Take the Massachusetts Institute of Technology (MIT) as an example. It produced the largest number of graduate scientists who received defense funding. 33% of those who graduated before 1946 are located in Superior, Wisconsin. From the perspective of a scientist who graduated from Michigan State University and arrived in Superior, being surrounded by MIT graduates in the city—due to the expansion of the scientific workforce from MIT driven by defense funding—can arguably be considered exogenous.

This approach parallels the shift-share instruments commonly used in the immigration literature, where the geographic distribution of immigrants is modeled as the historical settlement patterns of different national-origin groups interacted with contemporaneous immigration shocks (Hunt & Gauthier-Loiselle, 2010; Tabellini, 2020) and is inspired by recent literature in urban economics (Moretti, 2021; Giroud *et al.*, 2024). In a similar manner, I construct an instrument based on the historical share of graduates from each university who settled in a given city, interacted with postwar defense funding received by their university of origin. This interaction captures the extent to which defense funding indirectly influenced city-level scientist concentrations through pre-existing university-to-city connections, allowing me to isolate plausibly exogenous variation in city size. This approach builds on the intuition that defense funding is orthogonal to technological potential so that it can be used to estimate the effect of R&D on future technological change (Gross & Sampat, 2023; Moretti *et al.*, 2025). In many cases, defense funding is allocated based on strategic and political considerations rather than the intrinsic innovative capabilities of universities. Since defense priorities often reflect broader geopolitical imperatives instead of local technological potential, the variation in funding is plausibly exogenous to the latent determinants of technological progress. This exogeneity ensures that defense funding shocks can serve as a credible instrument for isolating the causal impact of R&D investments on subsequent technological change.

The validity of this empirical approach depends on whether the proposed instrument is both relevant for city size and uncorrelated with the outcome variables. Figure 2 presents supporting evidence. Panel (a) illustrates the relationship between the number of graduates from 1946 to 1970 and the number of scientists who received defense funding during that period, controlling for the number of graduates who completed their degrees before 1946 at the university level.

A more formal test of this relationship is provided in Table 1, which confirms a strong positive association between the magnitude of defense funding and the increase in the number of scientists. The relationship remains robust when controlling for the number of graduates before 1946, as well as for alternative instrumental variable candidates, such as funding for space, the NSF, atomic energy, and other research initiatives.

Panel (b) examines the relationship between the number of patents per individual scientist and the proposed instrumental variable at the city level. Specifically, it focuses on the number of patents granted between 1926 and 1940 for scientists who received their degrees before 1946. This result remains robust to the inclusion of additional controls, as shown in Table A.4.

3.1 Main Results

Figure 3 and Table 2 present the main results on the effects of city size on scientists productivity and wage. Table 2 reports both OLS and IV estimates, where the dependent variables are the number of patents granted between 1951 and 1976 and wages in 1970. The outcome variables are divided by their sample means so that the results can be interpreted as percentage changes relative to the mean.

OLS Estimates. Panel A of Table 2 presents the OLS estimates of the relationship between city size and the outcome variables. Across all specifications, the estimated coefficient on log city size is positive and statistically significant. In the baseline specification (Column 1), a 10% increase in city size is associated with a 1.3 percentage point increase in patent output. This effect remains robust after controlling for field fixed effects (Column 2), institution fixed effects (Column 3), and additional controls such as graduation year and gender (Column 4). Similarly, Columns 5-8 show that city size is positively associated with wages, with estimated coefficients ranging from 0.03 to 0.05 depending on the set of included controls.

IV Estimates. Panel B of Table 2 presents the instrumental variable (IV) estimates, using exogenous variation in city size driven by universities' propensity to receive defense funding. The first-stage F-statistics are consistently above conventional thresholds, indicating strong instrument relevance. The IV estimates suggest that the effect of city size on innovation is larger than the OLS estimates, with a 10% increase in city size leading to a 3.3 percentage point increase in patent output in the baseline specification (Column 1). This effect remains significant across alternative specifications, though its magnitude decreases to 1 percent when additional controls are included (Columns 3-4).

For wages, the IV estimates in Columns 5-8 indicate that city size has a positive and statistically significant effect, but the magnitudes are generally comparable to the OLS results. This suggests that the direct effect of agglomeration on wages is relatively stable across estimation methods. The elasticity of wages with respect to city size is very similar to the estimates of wage elasticity with respect to government R&D reported by Goolsbee (1998). Notably, Gruber *et al.* (2023) reports that the elasticity of patents and wages with respect to city size is 0.06 and 0.01, respectively—figures that closely align with the estimates in this paper. Carlino *et al.* (2007) estimates that the patent elasticity with respect to city size is about 0.2.

The larger IV estimates compared to OLS suggest that ordinary least squares may underestimate the effect of city size on innovation, potentially due to measurement error or endogenous sorting of scientists. The significant positive effect on non-funded scientists further supports the presence of knowledge spillovers. Meanwhile, the relatively stable wage effects imply that the benefits of city size extend beyond direct R&D funding to broader labor market dynamics.

To compare the value of increased invention with the corresponding rise in labor costs, I draw on the private value of patents estimated by Kogan *et al.* (2017), which suggests that the top 10 percent of patents were valued at $8,890,000 (in 1970 dollars). My results indicate that a 1 percent increase in city size leads to a 0.0028 increase in patents, which corresponds to an estimated increase of $24,892 in invention value (in 1970 dollars). At the same time, the estimated increase in wages due to the same 1 percent increase in city size is $448 per worker. These findings suggest that urban concentration fosters innovation at a rate that more than compensates for higher labor costs.

3.2 Agglomeration Effects and the Role of Direct Funding

Table 3 examines how the effect of city size—measured as the number of surrounding scientists—on individual productivity and wages varies depending on whether a scientist received direct defense funding. This analysis sheds light on the extent to which agglomeration effects, rather than individual subsidies, drive scientific productivity and wage.

The results indicate that the effect of city size on wages is relatively stable regardless of whether a scientist received direct funding. In Columns 3 and 4, the estimated coefficients on city size remain positive and statistically significant across both groups, suggesting that the wage premium associated with city size is not driven by direct government R&D support. Instead, it appears to reflect broader labor market forces, such as knowledge spillovers, labor demand effects, or competition among firms for skilled scientists.

In contrast, the effect of city size on patenting differs significantly depending on whether the scientist received funding. Columns 1 and 2 show that the positive effect of city size on patenting is more pronounced among scientists who did not receive direct defense funding. While the effect in Column 1 (which includes funded scientists) is small and not statistically significant, Column 2 (which excludes them) shows a positive and significant relationship between city size and patent output. This suggests that the productivity-enhancing effects of a larger scientific community primarily benefit those who do not receive direct government support.

These findings align with the idea that agglomeration effects and knowledge spillovers play a crucial role in fostering scientific productivity. Scientists who do not receive direct funding appear to benefit more from their surrounding environment, likely through informal knowledge exchange, collaboration opportunities, and shared resources. This provides further evidence that government R&D investment influences innovation not only through direct subsidies but also by shaping the broader scientific ecosystem.

4. Conceptual Framework

To assess the magnitude of the coefficients in my main findings and explore counterfactual scenarios, I use a simple economic model that formalizes the relationship between agglomeration, government funding, and scientific output. Building on David & Hall (2000), the model provides a theoretical foundation for estimating key structural parameters, enabling me to quantify the relative contributions of labor supply elasticity, knowledge spillovers, and congestion costs.

Model Setup. The model is defined as follows. Let G denote the public R&D budget, which is determined exogenously, and let L represent the total labor supply of R&D workers. The labor force is divided into private-sector R&D workers (L_P) and government-funded R&D workers (L_G), such that:

$$L = L_P + L_G.$$

The government budget constraint is given by:

$$G = wL_G,$$

where w denotes the wage rate of R&D workers.

The labor supply of scientists follows:

$$g(w) = L.$$

Labor demand is determined by equating the wage rate with the marginal product of private R&D labor:

$$w = K\big(bL_G\big)\, f\Big((1-b)L_G + L_P\Big),$$

where $b \in [0, 1]$ represents the share of funding allocated to basic research. The government assigns a fraction $(1 - b)$ of publicly funded researchers to applied research, while the remaining share is devoted to producing fundamental knowledge that enhances the productivity of all scientists, thereby shifting the labor demand curve outward. Here, K is a positive, concave function $(K' > 0, K'' < 0)$, and f is a continuous, monotonically decreasing function $(f' < 0, f'' < 0)$, ensuring that the labor demand curve is downward sloping.

Estimation. Solving the model yields the following conditions (derivations provided in Appendix B):

$$\frac{d\ln L}{d\ln w} = \eta \tag{3}$$

$$\frac{d\ln L_G}{d\ln w} = \frac{1-\psi\eta}{\phi} \tag{4}$$

$$\frac{d\ln L_G}{d\ln L_P} = \frac{(1-\psi\eta)\frac{L_P}{L}}{\phi\eta + (\psi\eta - 1)\frac{L_G}{L}} \tag{5}$$

Since these equations share common variables, I estimate them simultaneously using Seemingly Unrelated Regression (SUR). This approach allows me to efficiently estimate the key structural parameters—labor supply elasticity (η), the slope of the inverse labor demand curve (ψ), and the productivity parameter (ϕ)—while accounting for correlations in error terms across equations.

To implement this, I first run separate regressions for each dependent variable on the instrumental variable, which captures spatial variation in defense-related research funding. I then use the estimated coefficients to back out η, ψ, and ϕ through nonlinear transformations. The share of government-funded scientists in total scientific employment is approximately 0.1 (L_G/L), as reported in Table A.1.

Panel A of Table 4 presents the estimated results. The labor supply elasticity (η) is estimated to be 34.248, indicating that the supply of scientific labor is highly responsive to wage changes in the very long run.

The productivity growth from government-funded researchers is estimated to be 0.023% when there is a 1% increase in government-funded researchers. This estimate is close to Dyevre (2023), who finds that a 1% increase in government research budgets leads to a 0.017% increase in productivity.

The estimated slope of the inverse labor demand curve (ψ) is positive (0.029), implying that the labor demand curve is upward sloping. While standard labor demand models typically assume a downward-sloping relationship, an upward-sloping labor demand curve can emerge in the presence of strong agglomeration effects (Enrico, 2011). Nevertheless, I consider two alternative specifications in Panel B and C.

In Panel B, I assume a perfectly elastic labor demand curve, where $\psi = 0$, meaning that wages remain constant regardless of employment levels. Under this scenario, the estimated productivity parameter (ϕ) is 0.025, suggesting that a 1% increase in government-funded researchers leads to a 0.025% increase in productivity. This estimate is slightly larger than the baseline estimate in Panel A. Panel C considers the opposite case, where the labor demand curve is downward sloping with $\psi = -1$, meaning that an increase in employment leads to lower wages (Hamermesh, 1986). This specification results in a much larger estimated productivity effect, with ϕ increasing to 0.884%. This suggests that when labor demand is more constrained, the marginal impact of government-funded researchers on productivity is significantly stronger.

Counterfactual Scenario. The estimated results indicate that the labor supply of scientists is highly elastic in the long run and that government-funded research contributes significantly to productivity growth. However, recent evidence from Fernández-Villaverde *et al.* (2024) suggests that the increasing concentration of economic activity within large firms has led to greater labor market concentration, reducing the labor supply elasticity of scientists. To assess the potential impact of this change, I quantify how a decline in labor supply elasticity affects productivity growth.

To quantify the decrease in labor supply elasticity, I refer to Fernández-Villaverde *et al.* (2024), who document a 71% decline in labor supply elasticity between 1970 and

2019 using patent data. Applying this decline to my estimates, the labor supply elasticity is reduced to $\eta = 9.79$. Under these conditions, the estimated productivity growth from government-funded research is 0.004 with a standard error of 0.003 (p-value = 0.21), suggesting that the effect is no longer statistically distinguishable from zero.

These findings imply that declining labor supply elasticity may significantly weaken the productivity benefits of government-funded research. In an environment where scientists face fewer employment options due to firm consolidation, wages become less responsive to changes in demand, potentially dampening incentives for firms and institutions to expand research employment. This suggests that policies aimed at fostering competition in the scientific labor market, such as reducing barriers to mobility, increasing support for independent research institutions, or mitigating the effects of non-compete agreements, could help preserve the productivity gains from government R&D investments. More broadly, these results highlight how structural changes in labor markets can influence the effectiveness of public investments in science and innovation, underscoring the need to account for labor market frictions when designing policies to support research-driven economic growth.

5. Conclusion

This paper uses individual-level data on patents, wages, locations, and government funding status to estimate the impact of government research on scientists' productivity and labor costs. The findings show that labor supply elasticity is relatively high in the long run, and that government research funding generates positive spillover effects on scientists who do not directly receive funding but still experience productivity gains in their inventive activity. Moreover, the monetary value of these inventions exceeds the associated increase in labor costs, suggesting that government-funded research contributes to overall innovation growth.

The effect of government research on productivity is context-dependent. To better understand the underlying mechanisms, I extend an existing theoretical framework and identify labor supply elasticity as a key parameter shaping these dynamics. This revisits the insights of Goolsbee (1998), who emphasized the crucial role of labor supply elasticity in determining the effectiveness of government research investments.

Recent evidence suggests that labor supply elasticity may have declined in recent decades. Notably, the study period in this paper predates the main focus of Goolsbee (1998). If labor supply elasticity is lower than previously assumed, the productivity gains from government research would be mitigated. Ongoing research in this area explores various factors that may contribute to this decline. One possible explanation is rising market concentration, as documented by Fernández-Villaverde *et al.* (2024), which may limit employment mobility and reduce the responsiveness of labor supply. Another possibility is the burden of knowledge problem, where later generations of scientists must first absorb a growing body of existing research before making new discoveries (Jones, 2009; Bloom *et al.*, 2020).

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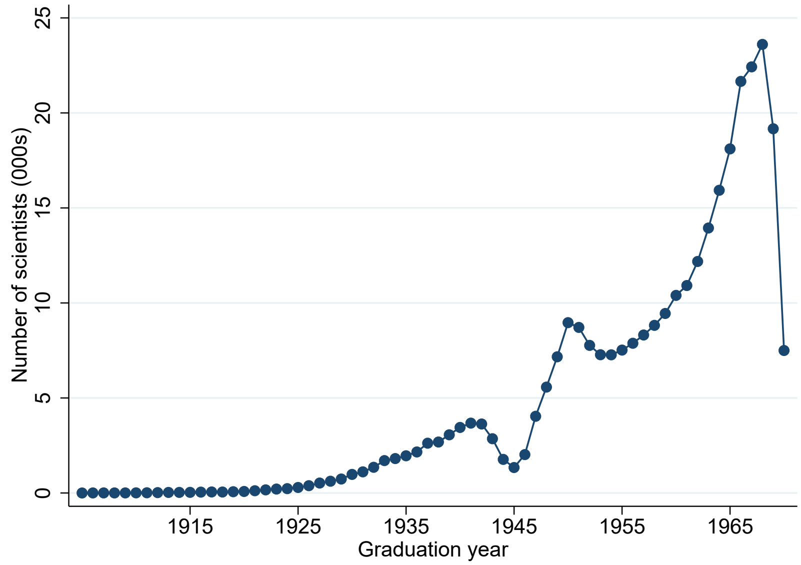
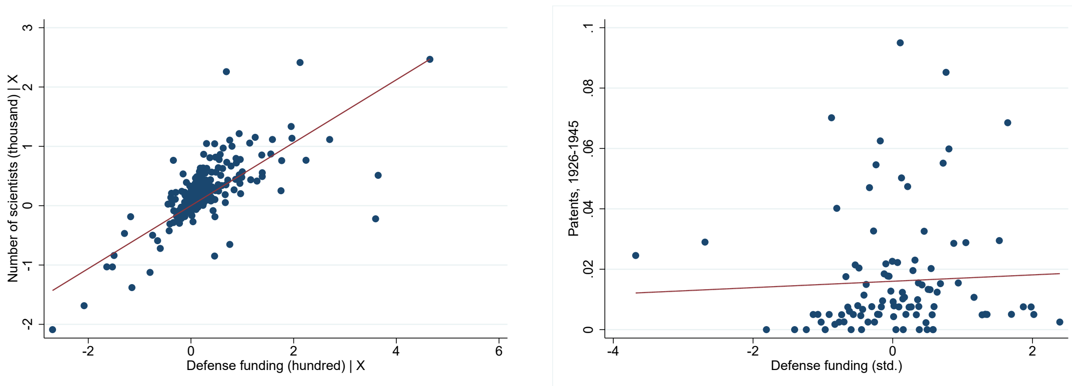
Figure 1: Number of scientists by graduation year



Note: These figures illustrate the number of scientists by graduation year in the National Register of Scientific and Technical Personnel Files (1970).

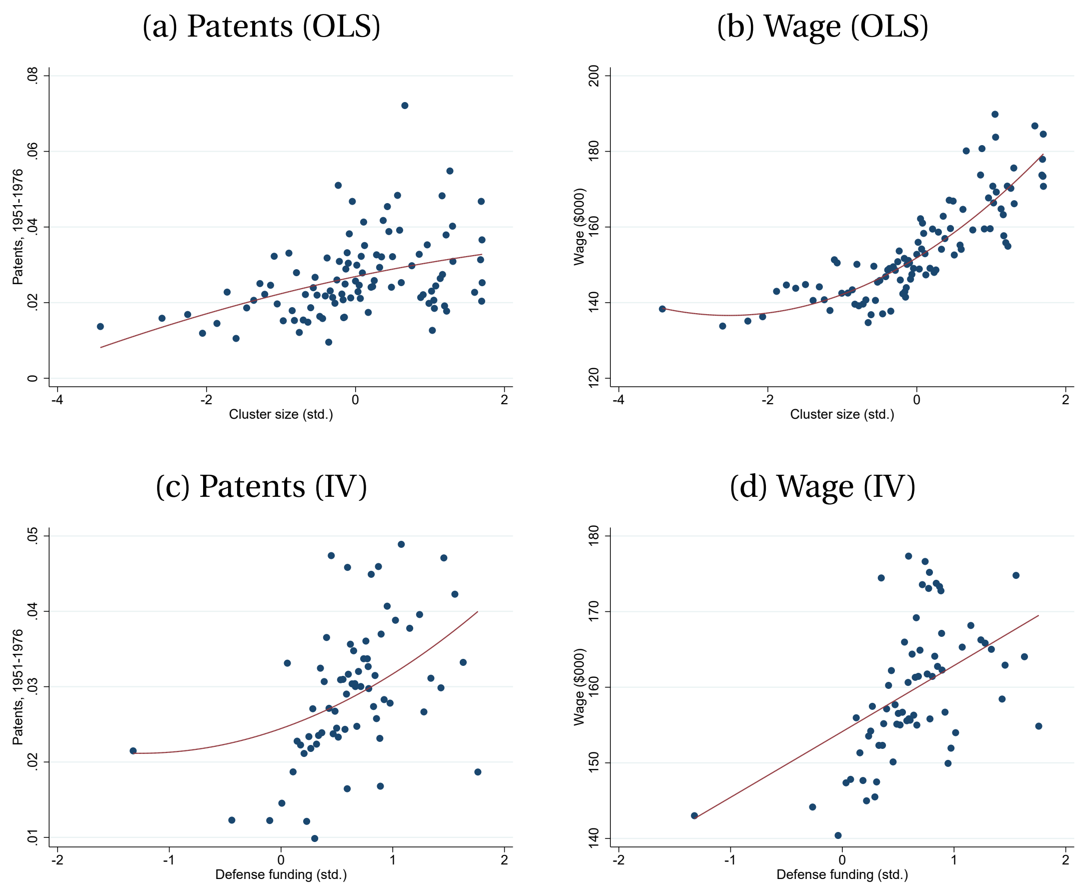
Figure 2: Validity of the instrument

(a) Number of scientists (b) Number of patents (placebo)



Note: These figures illustrate the relationship between defense funding, the number of scientists, and the number of patents during the pre-treatment period (1926-1945). The unit of analysis is a university in panel (a) and individual scientists in panel (b), presented as a binned scatterplot. The sample in panel (b) consists of scientists who received their degrees before 1946.

Figure 3: Number of AI papers and patents



Note: These figures present binned scatterplots illustrating the relationships between patents, wages, city size, and the instrumental variable. The dependent variables are indicated in the title of each panel. In panels (a) and (b), the independent variable is city size, while in panels (c) and (d), the independent variable is the instrumental variable. To facilitate interpretation, all independent variables are standardized. The lines represent a quadratic fit to the scatterplot.

Table 1: Zeroth stage: Defense funding and the number of graduates

DV:	Number of graduates, 1946-1970		
	(1)	(2)	(3)
Defense funding recipients	9.27***	5.30***	5.94***
	(0.63)	(0.68)	(1.57)
Space funding recipients recipients			-4.22
			(4.21)
NSF funding			3.22
			(7.73)
Atomic energy funding recipients			2.90***
			(0.98)
Number of graduates, pre-1946		2.58***	2.26***
		(0.27)	(0.28)
Observations	1,735	1,735	1,735
R-squared	0.86	0.93	0.93

Note: The unit of analysis is a university. Funding is measured as the number of students who received the respective funding in each university who graduated between 1946-1970, expressed in hundreds. Robust standard errors in parentheses. *** $p<0.01$, ** $p<0.05$, * $p<0.1$

Table 2: Effects of city size on patents and wage

DV:	Patents, 1951-1976				Wage, 1970			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: OLS estimates								
log city size	0.13***	0.10***	0.08***	0.08***	0.05***	0.05***	0.03***	0.03***
	(0.03)	(0.02)	(0.03)	(0.03)	(0.01)	(0.00)	(0.01)	(0.00)
Observations	276,588	276,588	272,666	272,666	235,451	235,451	232,120	232,120
R-squared	0.00	0.00	0.01	0.01	0.03	0.07	0.14	0.35
Panel B: IV estimates								
log city size	0.33***	0.31***	0.09*	0.11**	0.11***	0.10***	0.03***	0.03***
	(0.07)	(0.06)	(0.05)	(0.05)	(0.01)	(0.01)	(0.01)	(0.00)
Observations	276,588	276,588	272,666	272,666	235,451	235,451	232,120	232,120
R-squared	-0.00	-0.00	0.00	0.00	0.00	-0.01	0.01	0.01
KP F-stat	78.85	84.87	117.07	117.67	83.07	89.15	120.16	120.84
Field FE		Y	Y	Y		Y	Y	Y
Institution FE			Y	Y			Y	Y
Graduation year FE				Y				Y
Female FE				Y				Y

Note: Unit of analysis is individual scientist who received their degree between 1946 and 1970. The outcome variables in columns (1) to (4) represent the number of patents, while those in columns (5) to (8) correspond to wages. All outcome variables are normalized by the sample mean, allowing for interpretation in percentage terms. Standard errors are clustered by city. *** $p<0.01$, ** $p<0.05$, * $p<0.1$

Table 3: Agglomeration and the role of defense funding

DV:	Patents, 1951-1976		Wage, 1970	
	(1)	(2)	(3)	(4)
log city size	-0.07	0.12**	0.04***	0.03***
	(0.21)	(0.05)	(0.01)	(0.01)
Observations	25,825	246,595	24,064	207,805
R-squared	-0.00	0.00	0.01	0.01
KP F-stat	104.93	112.41	104.87	115.03
Defense funding recipient	Y	N	Y	N
Controls	Y	Y	Y	Y

Note: Unit of analysis is individual scientist who received their degree between 1946 and 1970. The outcome variables in columns (1) to (2) represent the number of patents, while those in columns (3) to (4) correspond to wages. All outcome variables are normalized by the sample mean, allowing for interpretation in percentage terms. All columns control for field, institution, graduation year, and female fixed effects. The samples in columns (1) and (3) consist of scientists who received defense funding from the government, while those in columns (2) and (4) include scientists who did not receive government funding. Standard errors are clustered by city. *** p<0.01, ** p<0.05, * p<0.1

Table 4: Estimated parameters

ϕ	ψ	η
Panel A: Upward Sloping Labor Demand		
0.023***	0.029***	34.248***
(0.006)	(0.009)	(11.676)
Panel B: Perfectly Elastic Labor Demand		
0.025***	0	
(0.007)		
Panel C: Downward Sloping Labor Demand		
0.884***	-1	
(0.057)		

Labor supply elasticity (η), the slope of the inverse labor demand curve (ψ), and the productivity parameter (ϕ). The labor demand elasticity in Panel A is estimated from the model, while in Panels B and C, it is calibrated. *** p<0.01, ** p<0.05, * p<0.1.

Appendix

A. Additional figures and tables

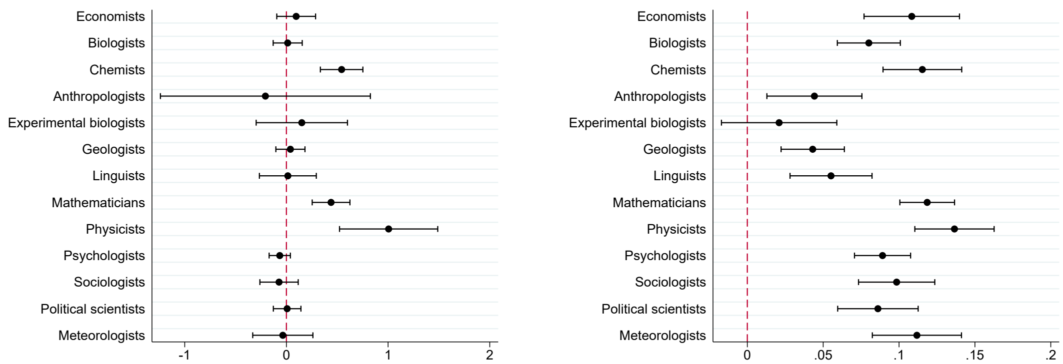
Table A.1: Distribution of Scientists by Government Defense Funding

Field	No Defense Funding	Defense Funding	Total
Economists	11,564	360	11,924
Biologists	42,002	1,162	43,164
Chemists	65,985	5,180	71,165
Anthropologists	1,132	15	1,147
Experimental Biologists	13,112	434	13,546
Geologists	19,249	1,335	20,584
Linguists	1,695	65	1,760
Mathematicians	29,983	6,111	36,094
Physicists	25,640	8,351	33,991
Psychologists	23,614	1,097	24,711
Sociologists	7,218	72	7,290
Political Scientists	5,815	258	6,073
Meteorologists	3,238	1,924	5,162
Total	250,247	26,364	276,611

Note: The table presents the number of scientists by field and indicates whether they received government defense funding. The sample includes scientists who earned their degrees between 1945 and 1970.

Figure A.1: Effects by subfield

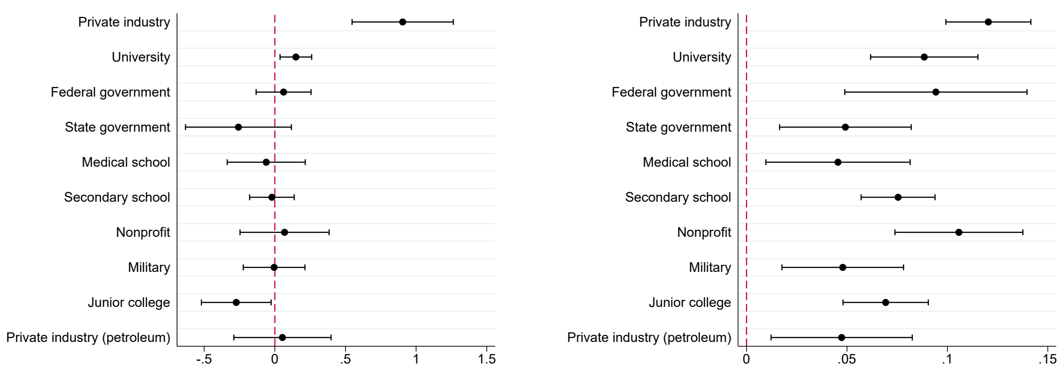
(a) Patents, 1951-1976 (b) Wage, 1970



Note: These figures present the estimated coefficients from the IV regression in Equation 1, showing the effect of city size on outcome variables for a subset of scientific fields.

Figure A.2: Effects by employer

(a) Patents, 1951-1976 (b) Wage, 1970



Note: These figures present the estimated coefficients from the IV regression in Equation 1, showing the effect of city size on outcome variables for a subset of the 10 most frequent employer type.

Table A.2: Correlation between defense funding and wage

DV:	Wage, 1970				
	(1)	(2)	(3)	(4)	(5)
Defense funding recipient	0.08***	0.08***	0.08***	0.06***	0.05***
	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Observations	235,465	235,465	232,134	232,134	232,111
R-squared	0.00	0.06	0.13	0.34	0.37
Field FE		Y	Y	Y	Y
Institution FE			Y	Y	Y
Graduation year FE				Y	Y
Female FE				Y	Y
SMSA FE					Y

Note: The unit of analysis is a scientist who earned a degree between 1946 and 1970. The outcome variable is the wage in 1970, which is normalized by the sample mean for interpretation in percentage terms. The independent variable is an indicator for whether a scientist received defense funding. Standard errors are clustered by city. *** p<0.01, ** p<0.05, * p<0.1

Table A.3: Effects of city size on patents (placebo)

DV:	Patents, 1926-1945			
	(1)	(2)	(3)	(4)
Defense funding (std.)	0.001	0.001	0.003	0.003
	(0.002)	(0.002)	(0.002)	(0.002)
Observations	39,889	39,889	38,778	38,776
R-squared	0.000	0.002	0.013	0.028
Field FE		Y	Y	Y
Institution FE			Y	Y
Graduation year FE				Y
Female FE				Y

Note: The unit of analysis is a scientist who received a degree before 1946. Outcome variables are the number of patents granted between 1926-1945. Standard errors are clustered by city. *** $p<0.01$, ** $p<0.05$, * $p<0.1$

Table A.4: Effects of city size on patents of different values

DV:	Top 10%	Top 5%	Top 1%	log citations
	(1)	(2)	(3)	(4)
log city size	0.108**	0.003**	0.005	0.005**
	(0.052)	(0.002)	(0.004)	(0.003)
Observations	272,666	272,666	272,666	272,666
R-squared	0.000	0.000	0.000	0.000
KP F-stat	117.67	117.67	117.67	117.67

Note: The unit of analysis is a scientist who received a degree between 1946-1970. The outcome variable in columns (1)–(3) represents the number of patents in the top 10%, 5%, and 1% by value, as defined in Kelly *et al.* (2021). The outcome variable in column (4) represents the number of citation-weighted patents ranked in the top 10% by value. All patents were granted between 1951 and 1976. Standard errors are clustered by city. *** $p<0.01$, ** $p<0.05$, * $p<0.1$

B. Derivation of Equilibrium Conditions

This section derives Equations 4 and 5. First, Equation 4 is derived by taking the first derivative of the labor demand curve:

$$w = K\big(bL_G\big)\, f\Big((1-b)L_G + L_P\Big).$$

Differentiating both sides,

$$dw = K' f b\, dL_G + K f'\big((1-b)dL_G + dL_P\big).$$

Rearranging terms,

$$dw = b(K'f - Kf')dL_G + Kf'dL_P + Kf'dL_G.$$

Using the definitions of ϕ and ψ,

$$dw = \frac{\phi w}{L_G} dL_G + \frac{\psi w}{L} dL.$$

Rewriting,

$$dw(1-\psi\eta) = \frac{\phi w}{L_G} dL_G.$$

Finally, dividing both sides by w and dL_G/L_G,

$$\frac{d\ln w}{d\ln L_G} = \frac{\phi}{1-\psi\eta}.$$

Next, Equation 5 is derived by noting that

$$\frac{dL}{L}\frac{1}{\eta} = \frac{dw}{w} = \frac{\phi}{L_G}dL_G + \psi\frac{dL}{L}.$$

Rearranging terms,

$$\frac{1}{\eta L}(dL_P + dL_G) = dL_G\left(\frac{\phi}{L_G} + \frac{\psi}{L}\right) + dL_P\frac{\psi}{L}.$$

Rearranging further,

$$dL_P(L_G - \psi\eta L_G) = dL_G(\phi\eta L + \psi\eta L_G - L_G).$$

Dividing both sides by $L_G dL_P/L_P$, we obtain

$$\frac{dL_G/L_G}{dL_P/L_P} = \frac{(1-\psi\eta)L_P}{\phi\eta L + \psi\eta L_G - L_G}.$$